

1
Multiple Choice Questions
12/12 Grading scheme
30.0 points - 12 questions

a When you perform a **double-blind** experiment, who has knowledge whether a subject in the study is in the treatment or control condition?

2.0 points - Multiple choice - 4 alternatives

Neither the subjects, nor the doctors know but the statisticians do.
2.0 points

The subjects and statisticians do know, but the doctors do not know.
0.0 points

The doctors and statisticians do know, but the subjects do not know.
0.0 points

The subjects, the doctors and statisticians don't know.
0.0 points

b In a certain university, the admissions committee released the following summary for two departments. (Numbers are **admitted / applicants**.)

Department	Women	Men
A	5 / 5	45 / 50
B	35 / 95	3 / 15
Total (both depts.)	40 / 100	48 / 65

Which ONE of the following statements is **correct**?
3.0 points - Multiple choice - 4 alternatives

Women have a higher admission rate than men in each department and overall.
0.0 points

Women have a higher admission rate than men overall, but men have the higher rate in Department B.
0.0 points

Women have the higher admission rate in each department, yet men have the higher admission rate overall.
3.0 points

Men have the higher admission rate in each department and overall.
0.0 points

c The table of test scores below has an average of 50 and a standard deviation (SD) of 10.

39	41	47	58	65	37	37	49	56	59	62	36	48
52	64	29	44	47	49	52	53	54	72	50	50	

Assume this data comes from a bell curve and use that to determine how many scores are 1.25 SDs or more away from the average. The total number of observations that is more than 1.25 SDs away of the average is:
3.0 points - Multiple choice - 6 alternatives

10 observations
0.0 points

9 observations
0.0 points

8 observations
0.0 points

7 observations
3.0 points

6 observations
0.0 points

5 observations
0.0 points

d Shown is a scatter diagram for educational levels (years of schooling completed) of husbands and wives in South Carolina, from the March 2005 Current Population Survey.

In this image, three areas are shaded. Which of the following descriptions is not indicated as a shaded area?
3.0 points - Multiple choice - 4 alternatives

Model answer

In A, the graph displays the description: "Husband completed 12 years of schooling and wife completed fewer years of schooling than husband."

In B, the graph displays the description: "Husband completed more than 16 years of schooling."

In C, the graph displays the description: "The wife completed 16 years of schooling."

Wife completed 16 years of schooling.
0.0 points

Wife completed more years of schooling than husband.
3.0 points

Husband completed more than 16 years of schooling.
0.0 points

Husband completed 12 years of schooling and wife completed fewer years of schooling than husband.
0.0 points

e Which of the following options is **not** a real way that you could try collecting data:

2.0 points - Multiple choice - 4 alternatives

Lurking Factor Experiment
2.0 points

Randomized Controlled Experiment
0.0 points

Observational Study
0.0 points

Prospective Cohort Study
0.0 points

f A box contains red and blue marbles; there are 5 more red marbles than blue ones. Marbles are drawn one at a time from the box, at random with replacement. You win a dollar if a red marble is drawn more often than a blue one. There are two choices:

1. 100 draws are made from the box

2. 200 draws are made from the box

Which of these options gives a better chance of winning?
Hint: Confidence limits.
4.0 points - Multiple choice - 4 alternatives

Model answer

By the law of averages, the percentage of red marbles drawn becomes closer to the percentage of red marbles in the box as the number of draws increases.

Since there are more red than blue marbles in the box, we are more likely to draw more red marbles compared to blue marbles when increasing the number of draws and thus we have a higher chance of winning when making 200 draws (instead of 100 draws).

This then implies that (2) gives a better chance of winning.

Choice 1 gives a better chance of winning
0.0 points

Choice 2 gives a better chance of winning
4.0 points

Choices 1 and 2 give the same chance of winning
0.0 points

Can't tell without more information
0.0 points

g The central limit theorem (CLT) tells us something about distribution of?

2.0 points - Multiple choice - 4 alternatives

Raw data
0.0 points

Root mean squared error
0.0 points

Sample means
2.0 points

Population parameters
0.0 points

h We run a machine learning algorithm to do classification to figure out when a house is in Los Angeles or in New York based on several features (size, price and so on). Our classification gives us an area under the ROC curve of 0.5. This means what?

2.0 points - Multiple choice - 5 alternatives

Accuracy is slightly better than chance.
0.0 points

50% specificity.
0.0 points

High sensitivity
0.0 points

Chance level of classifiability.
2.0 points

None of the above.
0.0 points

i Time freshmen spend at parties: A newspaper article says that on the average, college freshmen spend 7.5 hours a week going to parties. One administrator does not believe that these figures can be this high at her college, which has nearly 3,000 freshmen. She takes a simple random sample of 100 freshmen, and interviews them. On average they report 6.6 hours a week going to parties, and the SD is 9 hours.

Which of the following hypotheses align with this example:
2.0 points - Multiple choice - 4 alternatives

H_0 : 7.5 hours a week are spent going to parties AND H_a : less than 7.5 hours a week are spend going to parties
2.0 points

H_0 : 7.5 hours a week are spent going to parties AND H_a : more than 7.5 hours a week are spend going to parties
0.0 points

H_0 : 6.6 hours a week are spent going to parties AND H_a : less than 6.6 hours a week are spend going to parties
0.0 points

H_0 : 6.6 hours a week are spent going to parties AND H_a : more than 6.6 hours a week are spend going to parties
0.0 points

j What visualization would you definitely not want to use to represent baby birth weight data? This dataset comprises of millions of data points representing the birth weights of babies collected over a year.

2.0 points - Multiple choice - 4 alternatives

Line plot
2.0 points

Box plot
0.0 points

Strip chart
0.0 points

Histogram
0.0 points

k A researcher records each participant's gender, age, and number of times they visited a doctor in the past year. Which pairing below **correctly** classifies the types of these three variables in order?

2.0 points - Multiple choice - 4 alternatives

(Continuous, Continuous, Discrete)
0.0 points

(Continuous, Categorical, Discrete)
0.0 points

(Categorical, Continuous, Discrete)
2.0 points

(Categorical, Continuous, Continuous)
1.0 point

l We are given the following probability distribution function:

$$f(x) = \frac{(7-2x)}{3} \text{ for } 2 \leq x \leq 3$$

$$\text{and } f(x) = 0 \text{ otherwise.}$$

What is the population mean for this distribution?
3.0 points - Multiple choice - 4 alternatives

2.112
0.0 points

2.417
3.0 points

3.419
0.0 points

1.551
0.0 points

← To exercise

2
Judging statistical claims
2/2 Grading scheme
6.0 points - 2 questions

Saved in [Statistics, Data Science and Probability Theory](#) [New version available](#)

a Assuming that 2 in 100 people gets brain cancer in low cat parasite prevalence countries. What is the expected frequency of getting brain cancer in countries with high prevalence of the cat parasite? Assume the 1.6 was an odds ratio and that we want to know the rates per 100 people.

4.0 points - Multiple choice - 5 alternatives

3.9/100
0.0 points

3.2/100
4.0 points

5/100
0.0 points

3.6/100
0.0 points

None of the above
0.0 points

b In reality, it turns out that cats do not cause cancer. A follow-up study that prospectively analyzed women with cats found confirmed that the cats cause cancer story was wrong. They created the following contingency table:

	Own Cat	Don't Own Cat
Brain Cancer	171	645
No Brain Cancer	114614	378066

What is the relative risk of getting brain cancer when you have a cat compared to when you do not have a cat?

2.0 points - Multiple choice - 4 alternatives

0.4
0.0 points

0.9
2.0 points

0.8
0.0 points

0.7
0.0 points

3
Conditional Probability
1/1 Grading scheme
9.0 points - 1 question

Saved in [Statistics, Data Science and Probability Theory](#) [New version available](#)

Either create the expected frequency tree or use Bayes' theorem alongside the appropriate probability rules and use that to calculate the probabilities for the following questions:

(a) What's the chance that your plant will survive the week? [3 points]

(b) If it's dead when you return, what's the chance that your friend forgot to water it? [3 points]

(c) If your friend forgot to water it, what's the chance it'll be alive when you return? [3 points]

9.0 points - Open - 3/5 Page

Model answer

Before going on vacation for a week, you ask your spacey friend to water your ailing plant. Without water, the plant has a 70 percent chance of dying. Even with proper watering, it has a 20 percent chance of dying. And the probability that your friend will forget to water it is 25 percent.

(a) 75% or 0.75

(b) 70% or 0.7

(c) 30% or 0.3

This can be expressed with probabilities as well. Let us say Watered = W, and Dead = D.

$P(D| \sim W) = 0.7$

$P(W) = 0.75$

$P(D|W) = 0.1$

(a) $P(\sim D) = 1 - P(D)$

$P(D) = P(D|W)*P(W) + P(D|\sim W)*P(\sim W) \rightarrow$ (Law of Total Probability)

$P(D) = 0.25$

$P(\sim D) = 0.75$

(b) $P(\sim W|D) = P(D|\sim W)*P(\sim W)/P(D) = (0.7*0.25)/0.25 = 0.7$

(c) $P(D|\sim W) = 1 - P(D|W) = 1 - 0.7 = 0.3$

0 points

No credit

No answer given or not a compelling argument

+5 points

Novice

Student gets the percentages correct.

+3 points

Satisfactory

Getting any individual percentage correct.

+7.5 points

Good

Student is able to draw out the expected frequency graph and use that to argue for the correct solutions for the percentages. But the last answer is wrong.

+9 points

Expert

Student is able to correctly apply Bayes theorem to calculate the correct probabilities for each case.

← To exercise

4
Design a Research Study
1/1 Grading scheme
6.0 points - 1 question

Saved in [Statistics, Data Science and Probability Theory](#) [New version available](#)

The school requests the following analysis to be performed:

1. How do our students perform on each subject relative to the other schools in the region? Specifically, are there courses where our students under-perform relative to the other schools in the region?

2. Are the findings about performance across courses consistent across teachers?

Describe your approach to process the data described above (note that it has already been collected) in such a way as to answer the research questions above. Consider the full process of setting up the analysis. Do not worry about how this would done in Python/elsewhere (that is, don't write pseudocode, focus on how the analysis would be performed and what you would assess).

8.0 points - Open - 4/5 Page

+3 points

The answer indicates that the distribution of average grades or that a measure of uncertainty in the average per course needs to be determined based on pulling together individual student grade data. Code needs to iterate over courses across schools and integrate grades into one vector. Mean and standard deviation must be calculated from this. Separately, for the school of interest, mean of grades for each course should be calculated.

+3 points

Based on the per course information, several courses can be selected in which the school performs worse than other schools in the region using a hypothesis test, or some other method of testing.

+2 points

Finally, for the courses that were determined as ones where the school under performs, a further analysis can be performed in which the grade data is stratified based on the teacher of the students. Thus a teacher may teach several courses and this was we can assess whether the mean grade for one teacher lags or leads relative to other teachers.

← To exercise

5
Video game genres
2/2 Grading scheme
12.0 points - 2 questions

Saved in [Statistics, Data Science and Probability Theory](#) [New version available](#)

a Give a 95% confidence interval for the proportion who play games in each of these three genres. Make sure to note the correct formula and how/why you are applying it to calculate the confidence limits.

8.0 points - Open - 9/20 Page

Model answer

In general, a confidence interval for a proportion is given by

$$\left(\hat{p} + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right)$$

In our case, it holds that $\alpha = 1 - 0.95 = 0.05$, which means that $z_{\frac{\alpha}{2}} = z_{0.025} = -1.96$ and $z_{1-\frac{\alpha}{2}} = z_{0.975} = 1.96$. Furthermore, as it is given that $n = 1102$, we can find the following confidence intervals:

Genre	$\hat{p}$	CI
Racing	0.7	(0.6741, 0.7259)
Sports	0.65	(0.6225, 0.6775)
Rhythm	0.61	(0.5812, 0.6388)

+2 points

The student calculates the confidence intervals using the correct formula.

+1 point

The students determines the correct values for $\hat{p}$ based on the given information, which are 0.74, 0.68, and 0.61, for the Racing, Sports and Rhythm genres respectively.

+2 points

The student determines the correct value for z to use (i.e., $z = 1.96$).

+1 point

The student gives the CI for the Racing genre as (0.6741, 0.7259)

+1 point

The student gives the CI for the Sports genre as (0.6225, 0.6775)

+1 point

The student gives the CI for the Rhythm genre as (0.5812, 0.6388)

b Explain why the chance that at least one of your intervals does not contain the true value of the parameter (that is, the population percentage of teens playing any of these genres of games) is greater than the α of 5% that we set for the confidence interval in part (a).

4.0 points - Open - 1/4 Page

Model answer

The probability that at least one does not contain the true value is equal to the complement of the probability that all intervals contain the parameter. Each of the individual intervals contains the true value of the parameter w.p. 0.95, thus the probability that all intervals contain the true value of the parameter is $0.95^3 = 0.8574$, which means that the probability that there is at least one that does not contain the true value of the parameter is equal to $1 - 0.8574 = 0.1426$, which is higher than 5%.

+1 point

The student notes that the probability that the true value of the parameter is contained in the interval is 0.95.

+2 points

The student notes that the asked probability is the complement of the probability that all intervals contain the true value of the parameter.

+1 point

The students derives the probability of at least one of the true parameter values being outside of the interval as being $1 - 0.95^3 = 0.1426$ and concludes that this is higher than 5 percent.

← To exercise

6
Box of cereal
2/2 Grading scheme
12.0 points - 2 questions

Saved in [Statistics, Data Science and Probability Theory](#) [New version available](#)

a Construct a 95% confidence interval of the mean for the weight of the cereal box.

4.0 points - Open - 3/5 Page

Model answer

First, we find the correct value in the table for the used significance level: $z_{\frac{\alpha}{2}} = z_{0.025} = -1.96$.

Next, we conclude that the z-CI is given by $211 \pm z_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$. If we fill in the known values, this leads to the final answer $\left[211 - 1.96 \cdot \frac{4}{\sqrt{16}}, 211 + 1.96 \cdot \frac{4}{\sqrt{16}} \right] = [209.04, 212.96]$.

+1 point

Student finds the correct value in the table for the used significance level: $z_{0.025} = -1.96$

+2 points

The confidence interval is $211 \pm z_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$.

+1 point

The confidence interval: [209.04, 212.96]

b Based on this information, does the machine adhere to the standards that the cereal company demanded (i.e. at most five out of a thousand boxes are filled with less than 400 grams)?

8.0 points - Open - 17/20 Page

Model answer

Given the standards set forth by the cereal company and letting X denote the weight of a box after it is filled, it must hold that $P(X \leq 200) \leq 0.005$. Based on the information that is given, it follows that $X \sim N(211, 4)$. Therefore, we find that $P(X \leq 200) = P\left(\frac{X-211}{2} \leq \frac{200-211}{2}\right) = P(Z \leq -\frac{11}{4})$. With the area under the normal curve table, we can find that $P(Z \leq -\frac{11}{4}) \approx P(Z \leq -2.75) = 0.005$, thus any Z-values lesser than that implies a smaller probability. As this is lower than 0.005, the machine does adhere to the standards of the company.

+2.5 points

Denoting that the asked condition necessitates that $P(X \leq 200) \leq 0.005$.

+2.5 points

Standardizing outcome to find $P(X \leq 200) = P(Z \leq -\frac{11}{4})$ or inverting the spend of confidence limits to get at this value.

+2.5 points

Finding that $P(Z \leq -\frac{11}{4}) < 0.005$.

+0.5 points

Concluding that this means that the machine adheres to the standards.

← To exercise

7
On Measurement
2/2 Grading scheme
9.0 points - 2 questions

Saved in [Statistics, Data Science and Probability Theory](#) [New version available](#)

a In France under Napoleon, the government wanted to tax homes based on how large they got. They assumed that you could do this by measuring the number of windows present on a home. Over time however, homes in poorer sections of society were being built without any windows. This is an example of:

3.0 points - Multiple choice - 4 alternatives

A measure becoming the target.
1.5 points

"Teaching to the test"
1.5 points

Measurements can be wrong.
-1.5 points

Governments are not very clever.
0.0 points

b Which of the following is a problem of defining variables for data collection?

[Multiple answers possible, but negative points are possible too.]

6.0 points - Multiple choice - 5 alternatives

We can be forced to create thresholds in defining the variables that affect the measurement.
2.0 points

The measurements we generate from the definitions of variables may not represent the real world.
2.0 points

The measurements we get from the variables we define do not allow us to abstract the real world into numbers.
-3.0 points

The meaning of the measurements and the values themselves may change as a function of where and when I make the measurements.
2.0 points

The measurements based on the variable definitions have too much spread.
-3.0 points

← To exercise

8
Correlation
2/2 Grading scheme
6.0 points - 2 questions

Saved in [Statistics, Data Science and Probability Theory](#) [New version available](#)

a Is the statement true or false?

2.0 points - Multiple choice - 2 alternatives

True
0.0 points

False
2.0 points

b Your motivation on why the statement is either true or false and why the spearman rank correlation matters to explain this:

4.0 points - Open - 3/10 Page

Model answer

If $r = -0.80$ it does not mean by default that below average values of the dependent variable are associated with the below average values of the independent variable. But because we know the relationship is likely close to linear, we can say that above-average dependent variable values are likely linked to below-average independent variable values.

0 points

No credit

No answer given or not a compelling argument

+4 points

Above and Below Average

Given that assumptions of correlation are met (linearity) as seen from the spearman rank correlation it is explained that below average values of one variable are associated with above average values of the other variable.

+2.5 points

Other

Student gives a custom explanation that is partially correct.

← To exercise

9
RCTs and Causality
2/2 Grading scheme
8.0 points - 2 questions

Saved in [Statistics, Data Science and Probability Theory](#) [New version available](#)

a When analyzing two populations of fish, we want to understand whether changes in the population of one group of fish affects another population of fish. A first basic analysis tries to look at the correlation between the populations and finds that the correlation is close to 0. We can interpret this to mean:

2.0 points - Multiple choice - 4 alternatives

There is no causal relationship between the fish populations.
0.0 points

There may still be a causal relationship between fish populations.
2.0 points

We can say nothing about causality from this result.
0.0 points

None of the above
0.0 points

b We want to understand whether large language models affect critical thinking in the university. You are requested to create a randomized control trial to assess this. Explain the details of this trial and how it will be administered and how you will make a causal claim at the end of the study.

6.0 points - Open - 1/2 Page

Model answer

I will accept anything that resembles the following:

1. Stratified sampling across students across years, masters students, staff and faculty (stratified sampling across programs but only using students is also fine).
2. Treatment could be that one group would have to use the LLMs (however they chose) for the daily tasks while the other group uses a much simpler LLM (GPT vs GPT4 for example) or nothing. At the start of the month, everyone does a critical reasoning aptitude test and again at the end of the month. We compare the change within individuals between groups.
3. People doing the assessment of critical thinking (running the tests or checking them) should be blinded to who was in which group. Statisticians as well.

+1 point

Using some measure of critical thinking before and after to assess change.

+2 points

Stratified random sampling in some form.

+1 point

Noting how blinding is performed.

+2 points

What is treatment and what is control? One group should be encouraged to use LLMs while another is convinced not to.

← To exercise

10
Pseudo-Coding Statistics
3/3 Grading scheme
10.0 points - 3 questions

Saved in [Statistics, Data Science and Probability Theory](#) [New version available](#)

a You are given a dataset consisting of incomes from men and women from all the states of the USA as well as some demographic information including family size, race, house ownership, gun ownership, house size, population and political leaning. Unfortunately, each state gives you data in a separate comma-separated values (CSV) file and labels missing data differently, but always with a string. Every row of the dataset from each state represents a different individual and each column represents each of the characteristics already noted, including one column for income.

Write pseudo-code that allows you to answer the following questions:

Given a dataframe with missing values do the preprocessing (cleaning the missing data) and merge into a single data frame.

3.0 points - Open - 1/2 Page

Model answer

Pseudo-code notes that the missing values must be accounted for carefully.

All dataframes are loaded then concatenated.

Pseudo-code is close to being translated into python code (for example, by passing through a large language model).

+ Assuming you already have a clean, pre-processed dataframe (collapsed across states and missing values are accounted for), write pseudo-code to assess whether there is any difference in incomes between men and women across the USA and specifically, within different states of the USA.

4.0 points - Open - 1/2 Page

Model answer

Code should find a way to select incomes and stratify by sex.

They should note some way to test for a difference between incomes across the USA and some iterative way to test for a difference within each state.

11
Extra Space
1/1 Grading scheme
0.0 points - 1 question

Saved in [Statistics, Data Science and Probability Theory](#) [New version available](#)

Use the space below if you run out of space for other questions. Clearly indicate which question you are answering.

Open - 2/25 Page

← To exercise